

# Homework 4

Due Friday 05/23 at 5 p.m. PT

## Problem 1. Cart-Pole Racing. (30 points)

Consider the cart-pole system as depicted in Fig. 1, and described by the model:

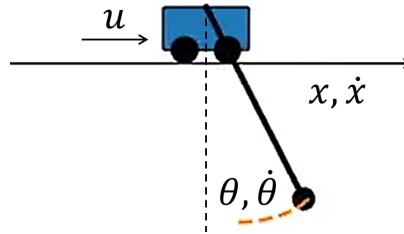


Figure 1: Schematic of a cart-pole system.

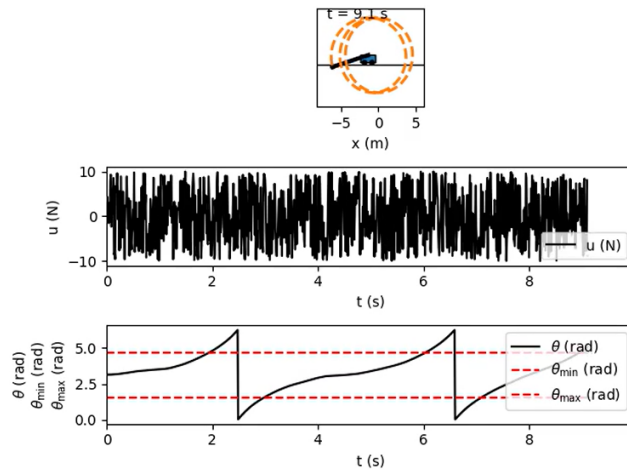
$$\frac{d}{dt} \begin{bmatrix} x \\ \theta \\ \dot{x} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{\theta} \\ \frac{m_p(l\dot{\theta}^2 + g \cos \theta) \sin \theta + u}{m_c + m_p \sin^2 \theta} \\ -\frac{(m_c + m_p)g \sin \theta + m_p l \dot{\theta}^2 \sin \theta \cos \theta + u \cos \theta}{l(m_c + m_p \sin^2 \theta)} \end{bmatrix},$$

with cart position  $x \in \mathbb{R}$ , pendulum angle  $\theta \in [0, 2\pi]$  rad  $\subseteq \mathbb{R}$ , cart velocity  $\dot{x} \in \mathbb{R}$ , and angular velocity  $\dot{\theta} \in \mathbb{R}$  defining the state  $\mathbf{x} = [x, \theta, \dot{x}, \dot{\theta}]^\top$ .  $\theta = \pi$  rad corresponds to the pendulum pointing straight upwards. The control consists of a force  $u \in [-\bar{u}, \bar{u}] \subseteq \mathbb{R}$  applied horizontally to the cart in the positive  $x$ -direction. The parameters include the mass of the pendulum  $m_p$ , the mass of the cart  $m_c$ , the length of the pendulum  $l$ , the gravitational acceleration constant  $g$ , and the maximum control effort  $\bar{u}$ . Set  $m_p = 1$  kg,  $m_c = 1$  kg,  $l = 1$  m,  $g = 2$  m/s<sup>2</sup>, and  $\bar{u} = 10$  N. **Note that we have set  $g = 2$  m/s<sup>2</sup>!** You can imagine that we have filled the pendulum with helium to make it easier for you to control the system.

We will hold a friendly competition with your classmates in a game of cart-pole racing! The rules of the game are simple. The system starts with an upright pendulum at  $\mathbf{x} = [0, \pi, 0, 0]^\top$ . A run is scored by the maximum distance that it achieves in the positive  $x$ -direction *before falling over*, with a maximum time limit of 10 s. Formally, we consider the pendulum to have fallen over if  $|\theta - \pi| \geq \pi/2$ . If the pendulum does not fall over within the time limit of 10 s, we say that the run has “qualified.”

1. The cart-pole system is 4D, however, we argue that you can conduct a formal safety analysis in a 2D space instead. Identify this 2D space and rigorously justify why this is allowed.
2. Compute the Hamilton-Jacobi (HJ) reachability value function in the 2D space using the tool of your choice. Please submit plots of the (time-varying) value function, as well as an analysis of your observations. **Please also include your code in your writeup submission.**
3. Navigate to `hw4/` in [this code repository](#). Follow the setup instructions in `README.md`. Check that your Python environment is configured correctly by running `python scripts/run_problem.1-3.py` to compete in the cart-pole race with a random controller. You should be able to generate an animation

as shown below.



The terminal should also print out the safety and performance scores for the run. Now, it's time to design and implement your own controller for competing in the cart-pole race! Implement the controller interface provided in `problem_1_3.py`. Compete with your controller by running `python scripts/run_problem_1_3.py`. If your controller involves randomness, please make sure to fix random seeds for reproducibility. Half of the points will be awarded based on your controller *performance*, up to a maximum score of 100 m. Half of the points will be awarded based on your controller *safety*, up to a maximum safe duration of 10 s. **Once you are done, please include all files necessary to run your code in your hw4.zip code submission. Also, include in your writeup an explanation of your controller design, the last frame of the simulation animation, and the scores from the terminal outputs.** *Hint:* If you are getting stuck, we encourage you to play around with simple hand-defined controllers to get a sense of the system dynamics. If you want to explore more advanced techniques, we suggest looking into methods like MPC, SCP, iLQR, and/or approaches that co-optimize performance and safety, as discussed in a guest lecture.

## Problem 2. Cart-pole Racing in a Windy Environment. (40 points)

Now, imagine that the cart-pole racing competition takes place outdoors, in a windy environment. There is an exogenous disturbance  $d \in [-5, 5]$  rad/s<sup>2</sup> in the  $\dot{\theta}$  dynamics:

$$\frac{d}{dt} \begin{bmatrix} x \\ \theta \\ \dot{x} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{\theta} \\ \frac{m_p(l\dot{\theta}^2 + g \cos \theta) \sin \theta + u}{m_c + m_p \sin^2 \theta} \\ -\frac{(m_c + m_p)g \sin \theta + m_p l \dot{\theta}^2 \sin \theta \cos \theta + u \cos \theta}{l(m_c + m_p \sin^2 \theta)} + d \end{bmatrix}.$$

You can assume that  $d$  evolves continuously and smoothly over time.

1. First, we look at a simpler setting where  $d$  is known to us beforehand to be a constant  $-1$  rad/s<sup>2</sup> (run A) or  $1$  rad/s<sup>2</sup> (run B). Implement the two controller interfaces provided in `problem_2.1.py`. The controllers can be different from each other, i.e., you can design them specifically for the known disturbances. Compete with your controllers by running `python scripts/run_problem_2.1.py`. Points will be awarded in the same manner as in the previous problem. **Please follow the same instructions from the previous problem for how to submit your work.**

2. Finally, consider the setting where  $d$  is unknown to us. The unknown disturbance can endanger the safety of your cart-pole, but it can also potentially help you achieve even better performance! Implement the controller interface provided in `problem_2_2.py`. The controller will be used across 5 different runs, named A, B, C, D, and E, in different disturbance conditions. Thus, the controller will need to adapt online using only the system history and without access to the underlying disturbances. Compete with your controller by running `python scripts/run_problem_2_2.py`. Half of the points will be awarded based on your controller *performance*, up to a maximum score of 200 m, 150 m, 100 m, 10 m, and 0 m for runs A, B, C, D, and E, respectively. Half of the points will be awarded based on your controller *safety* across the 5 runs, up to a maximum safe duration of 10 s for each run. **Please follow the same instructions from the previous problem to submit your work. Additionally, please follow the instructions in the Bonus Problem below to submit to the class leaderboard (under a pseudonym, if you like), even if you do not intend to compete for the top spots.**

**Bonus Problem. Climb the Cart-Pole Racing Leaderboard. (up to +30 points)**

The top 3 cart-pole racers [on the class leaderboard](#) will earn extra credit, based on their runs in Problem 2.2! To be considered for extra credit, all of your 5 runs must qualify, i.e., there must be no safety violation. Enter whether all of your runs qualify, as well as the average maximum distance your controller obtains across the 5 runs, [on this spreadsheet](#). **Even if you do not intend to compete for the top spots, please submit to the leaderboard so that everyone can get a sense of the state of the competition. If you choose to use a pseudonym, please specify it in your writeup.**